香 港 大 學 **MATH 2241**
THE UNIVERSITY OF HONG KONG **Assignment 3**

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Due Date: Monday, 5 October 2026, 22:00 pm

Instructions

Please read the format rules and the educational pilot notice below before completing your work.

1. **Assignment Format & Anonymity Requirements**
   - Write down your name, UID, and faculty clearly on this cover page only.
   - Please submit your assignment on Moodle.
   - **Do NOT write your name, UID, faculty, or any identifying marks on the interior solution pages (Page 3 onwards).** The internal pages are evaluated under randomized synthetic IDs to preserve anonymity during assessment.
   - Tablet users may annotate the assignment template directly with a stylus.
2. **Scope of the Educational Pilot**
   - This course utilizes **Co-Mark**, an assistive educational platform developed under an HKU student-staff partnership to support instructors with grading consistency, rubric alignment, and formative diagnostic feedback on complex coursework.
3. **Data Protection & Pseudonymization (PDPO Compliance)**
   - In accordance with the Hong Kong Personal Data (Privacy) Ordinance (PDPO), your Name and UID are kept on secure internal systems accessible exclusively by your course instructors, TAs, and authorized research staff.
   - Prior to script parsing, personal identifiers from this cover page are stripped and replaced with a randomized synthetic identifier (e.g., `Submission_ID: #8f2a1b`).
4. **Enterprise AI Infrastructure (Google Cloud Safeguards)**
   - Submissions are processed through commercial enterprise-tier Google Cloud APIs with strict data safeguards.
   - **Strictly No Model Training:** Commercial terms legally prohibit vendor use of your coursework, prompts, or feedback to train public AI models.
   - **No Human Review by Vendor:** Submissions are processed statelessly in memory and are never viewed by third-party vendor personnel.
   - **Secure Encryption:** Transmissions between our servers and the API are secured using industry-standard TLS encryption.

Instructions (Continued)

5. **Human Grading Authority & Intellectual Property**
   - **Assistive Only:** The AI generates draft diagnostics only; it **never** assigns official grades autonomously.
   - **Instructor Prerogative:** All official marks, point deductions, and feedback are reviewed and finalized exclusively by your human course instructors and TAs. Standard consultation and regrade rights remain completely unaffected.
   - **Intellectual Property:** You retain 100% of the copyright and intellectual property rights in your academic work.
6. **Data Retention & Voluntary Participation**
   - Identifiable records are retained strictly for the duration of the current academic semester and official grade appeal windows, after which personal identifiers are permanently decoupled from submission records.
   - Participation in the AI-assisted feedback pilot is voluntary. Submitting your coursework using this official assignment template implies agreement with these evaluation terms.

Question 1(a)(i)

Problem Context: Let (x_n) be a sequence in $[0,1]$. For each $n \in \mathbb{N}$, let $\epsilon_n > 0$. Suppose that

$$[0,1] \subset \bigcup_{n\in\mathbb{N}} (x_n - \epsilon_n, x_n + \epsilon_n).$$

Goal: The goal of this problem is to show that there exist finitely many indices $n_1, \ldots, n_k$ such that

$$[0,1] \subset \bigcup_{i=1}^{k} (x_{n_i} - \epsilon_{n_i}, x_{n_i} + \epsilon_{n_i}).$$

Part (a): Suppose, for contradiction, that no finite subcollection of these open intervals covers $[0,1]$, that is, there do not exist finitely many indices $n_1, \ldots, n_k$ such that

$$[0,1] \subset \bigcup_{i=1}^{k} (x_{n_i} - \epsilon_{n_i}, x_{n_i} + \epsilon_{n_i}).$$

Part (i): Explain how to construct a nested sequence of closed bounded intervals

$$I_1 \supset I_2 \supset I_3 \supset \cdots$$

such that no interval I_j can be covered by finitely many of the intervals $(x_n - \epsilon_n, x_n + \epsilon_n)$, and such that

$$|I_j| = \frac{1}{2^{j-1}}$$

for all $j \in \mathbb{N}$.

Base case : Let $I_1 = [0,1]$. By the assumption, I_1 can't be covered by finitely many of such intervals.

We assume that the statement holds for I_j, $j \le k$.

Inductive step : We divide I_k into $L: [a_k, \frac{a_k+b_k}{2}]$, $R\ [\frac{a_k+b_k}{2}, b_k]$.

Suppose both L and R can be finitely covered.

Then the union of two covers is still finite. So I_k also can be finitely covered, which is contradict to assumption.

∴ At least one of SL, RS can be chosen as I_{k+1}.

Thus, by induction, such intervals I_j holds.

Question 1(a)(i) (Continued)

Question 1(a)(ii)

For each $j \in \mathbb{N}$, choose a number $y_j \in I_j$ from the nested sequence of intervals constructed in part (i). Show that the sequence (y_j) is convergent.

The nested intervals $\{I_n\}_{n=1}^{\infty}$ satisfying: $|I_n| = \frac{1}{2^{n-1}}$

$\therefore \lim_{n\to\infty} |I_n| = 0$.

By the Nested Interval Property

$\bigcap_{j=1}^{\infty} I_j = \{y\}$, where $\forall n \in \mathbb{N}$, $a_n \le y \le b_n$.

Consider the two sequence: $\{a_n\}_{n=1}^{\infty}$ $\{b_n\}_{n=1}^{\infty}$.

Note that $a_1 \le a_2 \le \cdots \le a_n$, $b_1 \ge b_2 \ge \cdots \ge b_n$.

and $\forall n \in \mathbb{N}$, $a_1 \le a_n \le b_n \le b_1$

$\therefore$ They are both monotone and bounded.

By the Monotone Convergence Theorem, both sequences converge.

Let $\lim_{n\to\infty} a_n = A$, $\lim_{n\to\infty} b_n = B$.

$$B - A = \lim_{n\to\infty}(b_n - a_n) = \lim_{n\to\infty} |I_n| = 0 \quad \text{i.e. } A = B = y$$

$\because y_n \in I_n$

$\therefore a_n \le y_n \le b_n$

$\therefore$ By the Squeeze Theorem, $\lim_{n\to\infty} y_n = \lim_{n\to\infty} a_n = y$.

i.e. y_n converge to $y \in \bigcap_{n=1}^{\infty} I_n$

Question 1(a)(iii)

Use part (ii), or otherwise, to prove that there exist $m, N \in \mathbb{N}$ such that

$$I_N \subset (x_m - \epsilon_m, x_m + \epsilon_m).$$

$\because [0,1] \subset \bigcup_{n=1}^{\infty} (x_n - \epsilon_n, x_n + \epsilon_n)$

$\therefore \exists m \in \mathbb{N}.\ y \in (x_m - \epsilon_m, x_m + \epsilon_m)$. where y is the limit in part (ii).

Suppose $\forall N \in \mathbb{N}$, $I_N \not\subset (x_m - \epsilon_m, x_m + \epsilon_m)$.

then $\forall n \in \mathbb{N}$, $y \in I_N$ & $(x_m - \epsilon_m, x_m + \epsilon_m)$.

$\therefore |I_N| = \min\{ y - (x_m - \epsilon_m),\ x_m + \epsilon_m - y \}$ is fixed.

contradict to $\lim_{n \to \infty} |I_n| = 0$.

Question 1(b)

Use part (a) to derive a contradiction, and conclude that there exist finitely many indices $n_1, \ldots, n_k$ such that

$$[0,1] \subset \bigcup_{i=1}^{k} (x_{n_i} - \epsilon_{n_i}, x_{n_i} + \epsilon_{n_i}).$$

By the (iii), $\exists N, m \in \mathbb{N}$. $I_N \subset (x_m - \varepsilon_m, x_m + \varepsilon_m)$

which is a finite cover, contradict to (ii).

$\therefore$ the assumption in (i) drive a contradiction.

i.e. exist finite intervals, which union cover $[0,1]$

thus $\exists \{n_i\}_{i=1}^{K}$, $[0,1] \subset \bigcup_{i=1}^{K} (x_{n_i} - \varepsilon_{n_i}, x_{n_i} + \varepsilon_{n_i})$

Question 2

Prove or disprove: Every Cauchy sequence of rational numbers converges to a rational number.

False.

Consider $x_n = \frac{\lfloor 10^n \sqrt{2} \rfloor}{10^n} \in \mathbb{Q}$

$$\lim_{n\to\infty} \sqrt{2} - x_n = \lim_{n\to\infty} \frac{10^n\sqrt{2} - \lfloor 10^n \sqrt{2} \rfloor}{10^n} = 0$$

$\therefore \lim_{n\to\infty} x_n = \sqrt{2} \notin \mathbb{Q}$.

i.e. x_n is cauchy